Smart-X — Updates Based on Lecturer Feedback

Part 2 feedback

Compound blocks are split into one ledger row per distinct requirement.

Block A — Client UX

#	Lecturer point (quote)	Change made	Where in code	How to verify	Status
P2-1	“No state survives navigation.”	Added WASM singleton `OperatorSessionState`; Commands / Fleet / Device read and write filters and selections through it instead of page-local-only fields	`SmartX.Client/OperatorSessionState.cs`, `Program.cs` (`AddSingleton`), `Pages/Commands.razor`, `Pages/Fleet.razor`, `Pages/Device.razor`	On `/commands` set Health = Critical → open Overview (`/`) → return to `/commands`; Health filter still Critical	**Done**
P2-2	“Menu is a flat list of pages.”	Sidebar and Home regrouped by pillars: Telemetry · Command stream · Mesh routing (mesh unlocked in Stage 2)	`Layout/NavMenu.razor`, `Pages/Home.razor`, related CSS	Sidebar shows three labelled pillar groups with short hints; Home shows Pillar 1–3 cards and shortcuts	**Done**
P2-3	“No auto-refresh on the command stream.”	`PeriodicTimer` every 5 s on `/commands`; paused while override/undo runs; disposed with the page	`Pages/Commands.razor` (`AutoRefreshSeconds`, timer loop)	Stay on `/commands` without clicking Reload; status line “Auto-refresh every 5 s · last HH:mm:ss” advances	**Done**

Block B — Queues and priority

#	Lecturer point (quote)	Change made	Where in code	How to verify	Status
P2-4	“The queues are drained as they are filled. No background worker or separate consumer.”	Ingest/alert paths only enqueue and signal; `TelemetryQueueConsumer` (`BackgroundService`) dequeues priority then FIFO off the request thread	`GatewayStore.TryEnqueueOnly` / `RaisePriorityAlert`, `Storage/QueueWorkSignal.cs`, `Storage/TelemetryQueueConsumer.cs`, `Program.cs` hosted service	API console: consumer started log; after boot `GET /api/commands/snapshot` (or UI KPIs) shows `standardQueueDepth` ≈ 0 while `standardQueuePeakDepth` stayed elevated from the FIFO seed burst	**Done**
P2-5	“No priority tie breaker.”	Priority key `(severityRank, RaisedAtUtcTicks)` — equal severity → earlier `RaisedAt` first	`Storage/CriticalAlertQueue.cs`; proven by `Part2DemoSeeder.SeedPriorityTieBreaker`	API must boot successfully: seed asserts earlier Warning dequeues before later Warning of the same severity	**Done**

Block C — Disconnects

#	Lecturer point (quote)	Change made	Where in code	How to verify	Status
P2-6	“No disconnect detection.”	Freshness-based detection: refresh last-seen, raise disconnect alerts for disconnected nodes, clear keys on recovery; consumer polls regularly; HashSet still dedupes	`GatewayStore.DetectAndRaiseDisconnects`, `TelemetryQueueConsumer`, `Part2DemoSeeder.SeedDisconnectDuplicateSuppression`	Unique keys include `disconnect:sx-env-sump-temp`; feeder Stale nodes no false disconnect keys; second raise of the same key is suppressed	**Done**

Block D — Suggestions

#	Lecturer point (quote)	Change made	Where in code	How to verify	Status
P2-7	“Uses counts, not confidence. Fixed rules dominate — the moisture rule is hard-coded.”	Suggestions expose normalised `Confidence` in 0, 1; moisture→pump hard-code removed; follow-ups come from learned sequence pairs / pattern scoring	`SuggestionItem.Confidence`, `Storage/RecommendationEngine.cs`, `Pages/Commands.razor` Suggested Actions UI	UI/API show confidence `0.00–1.00`; no hard-coded title “Moisture fault → consider sump pump”; `StartPump` can still appear via pattern seeds	**Done**
